

Precision biomarkers of the exercise-induced immune “open window”: a scoping review of the validation readiness of conventional and omics candidates

Protocol version: 3.2 — amendments PRE-007 and PRE-008 of 6–8 October 2026 to version 3.1 (archived 5 October 2026 as GitHub Release v3.1, Zenodo version DOI 10.5281/zenodo.23163768)

Status: Version 3.2 archived (8 October 2026; version DOI 10.5281/zenodo.23229246). The 50-record screening pilot and the 10-report charting pilot have been completed and accepted as calibration; amendment PRE-004 revises the charting rules and two screening rules, amendment PRE-005 introduces two-stage screening (title, then title/abstract), and amendment PRE-006 sets the databases to PubMed, Web of Science Core Collection and Scopus with a Google Scholar supplement, adds search-stage limits and adopts the single-route strategy v0.9 (E, L). The PubMed export of 5 October 2026 under strategy v0.7 is superseded. The formal database searches were run on 5 October 2026 (PubMed, Web of Science Core Collection, Scopus). Amendment PRE-007 (6 October 2026, registered retrospectively for the design A adopted on 5 October 2026) introduces AI-stand-in first-pass screening with mandatory human verification: stage-1 title screening of all 15,750 deduplicated records was executed on 5–6 October 2026 (7,606 advanced) and stage-2 title/abstract screening of 9,277 retained records was completed on 7 October 2026 (4,326 advanced). Amendment PRE-008 (8 October 2026) triages the stage-2 advances by AI element extraction with an adjudicating read and layered human verification: only the validation-readiness layer proceeds to full text, and the evidence-map layer is charted from abstracts (C, F, I). PRE-007 and PRE-008 are released together as version 3.2, before human verification and full-text screening start. Amendment PRE-009 (8 October 2026) replaces independent dual human extraction with AI-assisted extraction and human verification (G.1), and amendment PRE-010 (8 October 2026) replaces dual independent human full-text screening with AI pre-filled full-text screening and human override (F); both are adopted and will be released as version 3.3, with an implementation update of PRE-004, before full-text screening starts. Human verification of the AI passes, full-text screening, extraction and synthesis have not started, and no formal included study or result exists.

Target journal: *Frontiers in Immunology*, under the Systematic Review article type, which includes scoping reviews; specialty section to be selected. Submission will include the PRISMA-ScR checklist and flow diagram as supplementary material and an AI-use disclosure in both Methods and Acknowledgments stating tool name, version, model and source.

Registration and archiving: Public GitHub repository with GitHub Releases archived by Zenodo. Version 3.0 was archived on 5 October 2026 (version DOI 10.5281/zenodo.23147973; concept DOI 10.5281/zenodo.23147972); version 3.1 was archived on 5 October 2026 (version DOI 10.5281/zenodo.23163768) under the same concept DOI; version 3.2 (bundling PRE-007 and PRE-008) was archived on 8 October 2026 (version DOI 10.5281/zenodo.23229246) under the same concept DOI. The project has never been registered in any registry; PROSPERO does not accept scoping reviews and OSF Registries is no longer the planned route.

Team: Access to PubMed, Web of Science Core Collection, Scopus, Google Scholar and full texts, and a second human reviewer, are confirmed. Team members are identified by codes: A (principal investigator, independent PRESS reviewer, analysis and writing lead), B and C (independent reviewers), D (search lead, data manager, adjudicator). Funding and competing-interest declarations remain to be supplied.

This is the canonical English protocol. The Chinese protocol is an aligned implementation version. The rendered English source and PDF are generated from this complete document; they are not a shortened independent protocol. Operational appendices in 03_search–06_synthesis implement the decisions below. This version follows three pre-archive activities disclosed in J: an internal protocol (version 1.0, January 2026), public exploratory reconnaissance and source verification on 2 October 2026 (Codex), and a collision re-check of the narrowed framing against published and registered reviews on 4 October 2026 (Claude Code;

02_preliminary/collision_recheck_2026-10-04.md). The protocol date is not a completed formal search cutoff. No search yield, eligibility decision, included-study count or review result is claimed here.

A. Background and rationale

Exercise can alter circulating immune-cell counts, immune mediators, mucosal measurements and immune-cell function over different time scales. The “open window” has been used to describe the hours to days after strenuous or prolonged exercise during which immune changes have been proposed to coincide with reduced host protection [15,16]. Whether such a period exists remains debated: lymphocytopenia may reflect redistribution rather than loss of whole-body function, and plasma-volume change, circadian timing and sampling alter interpretation [1,2]. This review does not adjudicate that debate. It uses the open window solely as an organizing construct that defines the post-exercise period (0–72 hours after cessation of an identifiable bout), the sampling time points and the marker families within which candidates are charted, irrespective of whether original authors endorse, reject or omit the hypothesis.

Candidates are charted by marker family. Conventional families include leukocyte and lymphocyte redistribution, natural-killer-cell counts and cytotoxicity, neutrophil function, T-cell subsets and function, mucosal secretory IgA, mucosal antimicrobial proteins such as lactoferrin and lysozyme, cytokines and other inflammatory mediators, and immunoglobulins and complement. Omics families include immune-cell transcriptomics, proteomics, metabolomics and lipidomics, epigenomics, non-coding RNA and single-cell measurements. Higher measurement dimensionality does not alone confer immune specificity or individual predictive value. A group-average response, an in-silico pathway annotation, an association with symptoms and a validated individual prediction address different questions. Candidate usefulness also depends on the exercise exposure, tissue or fluid, analytical method, sampling interval and intended monitoring use.

The intended contribution is a per-marker **validation-readiness map** of candidates measured within the organizing window. Throughout this protocol and its appendices, a validation-readiness map means a descriptive, domain-by-domain profile, for each marker or marker family, of which of the seven precision-evidence domains defined in C have actually been investigated and with what result. It is not a score, ranking, grade or development ladder, and every later prohibition on readiness scores or rankings refers to this definition. The map emphasizes individual baselines, within-person variation, repeatability, time-resolved interpretation, functional or clinical links and independent validation. It is not a catalogue that labels every statistically significant omics feature a precise biomarker. Analytical and clinical validation must be interpreted for a specified context of use [3].

In their 2017 review of immune recovery after exercise, Peake and colleagues concluded that, “except for salivary IgA, clear and consistent markers of this immunodepression remain elusive” [15]; that statement is the auditable starting point for this map. The debate on whether the open window exists is held by Peake et al. 2017 [15], Campbell and Turner 2018 [1], Simpson et al. 2020 [16], Reitzner and Brodin 2026 [19] and Xie et al. 2026 [17]; this review does not join it. Narrative reviews have also proposed marker agendas around the open window: Xie et al. 2026 in *Frontiers in Immunology* (tissue-resident memory T cells and their circulating precursors) [17]; Shi et al. 2025, also in *Frontiers in Immunology* (workload-dependent recovery of conventional indices) [18]; Reitzner and Brodin 2026 (the immune system as a systems-level biosensor of health and performance) [19]; and Mănescu et al. 2026 (molecular biomarkers of training responses) [20]. Davison et al. 2025 questioned the clinical relevance of ex vivo immune assays [2]. None of these charts human evidence marker by marker against prespecified validation domains with a reproducible search, and systematic reviews to date address single marker families or single populations: exercise-associated non-coding RNAs, B-cell-related immunity, TH1/TH2 markers in older adults and exercise immunometabolomics [4–7]. The registered reviews nearest to this protocol

differ in Population–Concept–Context: PROSPERO CRD42020186264 is confined to salivary IgA and respiratory-infection risk; PROSPERO CRD42022324139 is confined to the acute effect of high-intensity interval training; and OSF osf.io/4hsdr shares the JBI and PRISMA-ScR method stack but is restricted to athletes aged 10–17 years and to conventional markers, without omics, open-window or validation dimensions, whereas this review includes adults only. The 4 October 2026 re-check that identified these items is filed at `02_preliminary/collision_recheck_2026-10-04.md`. We make no “first review” or “no competing review” claim. The proposed distinction is the joint, time-resolved charting of conventional and omics candidates within one organizing window against separately recorded precision domains. That distinction will be rechecked against published and registered reviews before the v3.0 archive release and at the final search update.

A scoping review is appropriate because the primary questions concern the distribution, measurement, interpretation and validation of heterogeneous evidence, rather than one intervention effect or diagnostic threshold. We will follow JBI scoping-review methods [8], report with PRISMA-ScR [9], and report searches with PRISMA-S [10]. Prespecified, design-appropriate critical appraisal will support interpretation. We will not conduct a meta-analysis or turn appraisal into a clinical ranking.

B. Objective and review questions

This scoping review will map human evidence on candidate immune biomarkers measured within the exercise-induced “open window”, used here solely as an organizing construct that defines the post-exercise period after an identifiable bout, the sampling time points and the marker families to be charted. Across conventional immune indices and omics candidates in generally healthy adults after single identifiable bouts (Core A) and across repeated bouts in the same individuals (Support B), we will chart, marker by marker, which of seven precision-evidence domains have actually been investigated: analytical reliability, temporal validity, immune specificity, individualization, functional or clinical linkage, independent validation and use, and feasibility. The resulting per-marker validation-readiness map is a descriptive domain-by-domain profile, not a score, ranking or development ladder. The review does not test whether an open window of increased infection susceptibility exists, and original authors’ endorsement or rejection of that hypothesis neither determines eligibility nor weights evidence.

1. **Measurements and context:** Which marker families, individual biomarkers, analytical platforms, immune compartments and sample matrices have been measured within the organizing window (0–72 hours after cessation), in which adult populations and exercise settings?
2. **Timing, confounding and interpretation:** Which time points within the window (immediate, 30 minutes to <3 hours, 3 to <24 hours, 24 to 72 hours) and which outside-window recovery points are covered; how are cell composition, hydration, sampling and other relevant confounders addressed; and how do original authors interpret their findings in relation to the open window (charted descriptively only)?
3. **Individualization:** What evidence exists on personal baselines, within-person variability, repeatability across bouts, response heterogeneity, subgroup interpretation or individualized models, and how has it been validated?
4. **Function, outcomes and use:** Which functions or clinical outcomes were merely co-measured, which were associated with biomarkers, and which monitoring or prediction uses underwent independent validation? Together with questions 1–3 this yields the per-marker validation-readiness map and the development requirements that remain.

Positive, null and inconsistent findings are all relevant. The questions do not presuppose that exercise is immunosuppressive, nor test whether an open window exists, nor assume that any candidate is ready to predict infection.

C. Framework and operational definitions

Eligibility uses Population–Concept–Context (PCC). “Post-exercise” begins at cessation of an identifiable bout or competition. Exact times and their reported origin are retained. During-exercise samples can be ancillary observations in an eligible study but cannot alone establish post-exercise eligibility.

The **organizing post-exercise window** is the period from cessation to 72 hours inclusive. It is a charting frame, not an eligibility cap: a study qualifies through at least one post-cessation immune-related sample, and samples taken after 72 hours in an otherwise eligible study are extracted in full and tagged “outside-window recovery” because they document recovery endpoints. No exclusion code refers to the window. **Author open-window interpretation** is a descriptive tag recording whether the original authors used the open-window term, invoked immunosuppression or susceptibility without the term, interpreted changes as redistribution, or offered no window interpretation; it affects neither eligibility nor weighting.

For this project, **precision monitoring** means interpreting an immune-related measurement within the organizing window for a defined person or subgroup, exercise context, sampling time and intended monitoring use. Seven domains are recorded separately:

Domain	Evidence to chart
Analytical reliability	Assay performance, limits, quality control, reproducibility and pre-analytical handling
Temporal validity	Exact exercise-to-sample intervals, position relative to the organizing window, longitudinal coverage and how response or recovery is defined
Immune specificity	Direct immune measurement, cellular origin, composition adjustment and relevant alternative explanations
Individualization	Personal baselines, within-person noise, repeated-bout reproducibility, response heterogeneity, subgroup or individual models
Functional or clinical linkage	Direct immune-function measurements, tested associations and explicitly defined clinical outcomes
Independent validation and use	Technical confirmation versus independent participants/cohorts, model validation and tested decision use
Feasibility	Sampling burden, accessibility, cost, turnaround and field implementation, when reported

These are project operational domains, not a validated scale, regulatory designation or ordinal development ladder. There will be no combined “precision score”; domain evidence is summarized only as the per-marker validation-readiness map defined in A. Omics measurement, machine learning or a group mean difference alone does not establish personalized usefulness. Individualization or clinical validation is **not required for inclusion**; absence of such reporting is itself a mapping finding.

The **validation-readiness element** (amendment PRE-008) is the abstract-level signal used to triage stage-2 advances; it is a triage device, not an eligibility criterion or a precision-domain judgment. It is present when the title or abstract shows at least one of four sub-types. *Metric*

validation: a reported reliability, reproducibility or within- or between-person variability statistic, a reference interval, a derived or tested threshold, or a diagnostic or predictive performance measure; a remark that responses varied is not enough. *Outcome linkage*: an infection, upper-respiratory, illness, overtraining or performance outcome that was measured in the study and analysed against the marker; background framing, hypotheses or discussion do not count. *Omics discovery*: proteomic, metabolomic, transcriptomic, miRNA, epigenomic, single-cell or extracellular-vesicle profiling of immune-relevant candidates. *Repeated monitoring*: the same marker tracked in the same people across three or more identified bouts, or across a season, training block or multi-week period with a monitoring purpose; crossover or randomised comparisons of exercise conditions do not count, although they can qualify as Support B.

A **response** is an observed post-exercise measurement or contrast. A **recovery trajectory** requires longitudinal observations and an explicit endpoint; a single early sample or a nonsignificant comparison with baseline does not establish recovery. Cell-count normalization, biomarker normalization, immune-function recovery and clinical recovery remain distinct.

Immune linkage requires a direct immune-cell/count/subset/effector measure; a recognized immune mediator such as an immunoglobulin, cytokine or complement component; omics measured in immune cells; or biofluid omics with an immune objective or analysis stated in the original aims/methods and identifiable immune candidates. Original-study preregistration is not required. Mere post-hoc immune enrichment in an otherwise general muscle or metabolic study is insufficient for the main map. Cortisol, creatine kinase, lactate or heart-rate variability alone do not qualify, but may be charted as covariates in otherwise eligible studies.

Direct functional assays are distinguished from computational annotation. Co-measurement of a function or clinical outcome does not establish a biomarker–outcome association. Self-reported symptoms, clinician-defined syndromes and pathogen-confirmed infection are separate outcome classes. Association, temporal prediction, calibration, discrimination and clinical utility are recorded separately. We use “prediction” for future outcome prediction here, without assuming the treatment-response meaning of a predictive biomarker in other taxonomies.

Missing information is coded **NR** (not reported), **NA** (not applicable) or **UNCLEAR** (information present but insufficient or ambiguous after checking). NR is not evidence of biological absence or methodological failure.

D. Eligibility criteria

D.1 Population

Eligible participants are adults aged 18 years or older, of any sex or training status, who are generally healthy at baseline and do not have an active infection. Athletes, recreational exercisers and untrained participants are eligible; training strata are charted separately. Older adults are eligible and flagged as a subgroup. Children and adolescents (<18 years) are excluded; mixed-age cohorts are eligible only where an adult stratum is separable in the report. A cohort, or each separately reported group, counts as adult when an explicit age range or minimum age of at least 18 years is reported, or when the mean minus two standard deviations is at least 18 years; descriptors such as “adults” or “students” do not suffice. Otherwise the report remains awaiting classification, not excluded, unless it states that participants younger than 18 years were included (amendment PRE-004). This boundary distinguishes the review from the registered youth-athlete scoping review osf.io/4hsdr (10–17 years). Extractable healthy adult strata within cohorts of mixed health status are also eligible. Cohorts with known clinical/metabolic disease or disease-treatment exercise programmes are excluded from the main map. Ambiguous metabolic-health descriptions, including insulin resistance without a clear clinical classification, require explicit adjudication using the reported diagnosis and separability of strata, rather than an automatic healthy designation.

D.2 Exercise context and evidence streams

Core A — acute challenge: An identifiable single bout or competition of endurance, resistance, interval/sprint or mixed exercise, with at least one post-cessation immune-related sample and either a pre-bout baseline or a suitable non-exercise/comparator condition. No intensity, duration or mode threshold is imposed: any identifiable bout qualifies, and intensity, duration and mode are recorded as reported and used as evidence-map strata, never as eligibility filters, because the open-window literature defines “prolonged” and “intense” inconsistently and a threshold would discard comparator evidence. Immediate samples are eligible and there is no minimum 30-minute follow-up. The organizing window of 0–72 hours after cessation frames the charting; it is not a maximum interval, and samples beyond 72 hours in an eligible study are extracted and tagged “outside-window recovery” (C). During-only studies are ineligible. Repeated acute challenges before and after a training programme qualify for A; training moderation is extracted separately. Immediate-only studies contribute response evidence, not a complete recovery trajectory.

Support B — repeated post-exercise windows in the same people: The same immune-related marker is measured in the same individuals in the post-exercise windows after at least two identifiable bouts or competitions, with a known or reliably recoverable exercise-to-sampling relation and a baseline or appropriate comparator enabling interpretation. B is the cross-bout monitoring layer and the main evidence source for the individualization and repeatability domains; it is presented separately from A. A cohort may meet both A and B; linked cohort identifiers prevent duplicate overall counting. General weekly or seasonal resting surveillance with unknown last-bout timing, training pre/post snapshots with unknown exercise-to-sampling intervals, and habitual-exercise cross-sectional comparisons are background only: when such a report is otherwise an original study of generally healthy adults with an eligible immune link, it is retained at full text under the administrative status `RETAIN_BACKGROUND` and charted at reduced depth (G.1). A long observation period cannot substitute for an identifiable post-exercise time relation.

Studies involving supplementation, diet or other co-interventions qualify only where an exercise contrast is interpretable separately, for example through an appropriate control/placebo arm or factorial contrast. Otherwise the inseparable intervention is ineligible. Prior training, co-interventions and recovery arrangements are extracted.

D.3 Concept, design and publication status

Eligible studies meet the immune-link definition in C and report original human measurements. There is no requirement for the words “precision,” “biomarker,” “open window” or “recovery,” for a particular direction or significance level, or for a functional/infection endpoint. Studies are neither selected nor weighted according to whether their authors endorse, reject or ignore the open-window hypothesis; the author interpretation is charted as a descriptive tag (C). Designs may include randomized or nonrandomized experiments, crossover studies, before–after studies and observational designs that satisfy the timing and comparator rules. No arbitrary minimum sample size is imposed; small designs are clearly identified and appraised appropriately.

The primary A/B map includes peer-reviewed original reports. Reviews inform the rationale and citation chasing; they are excluded from the map (FT01), not retained as background records. In the database searches, reviews are removed by a publication-type limit and kept as a logged list of citation-chasing sources (E). Animal-only or in-vitro-only studies, protocols, editorials, conference abstracts without a full original report and pure methods papers without eligible human exercise observations are excluded from the primary map; potentially relevant abstracts are tracked for full publications.

A bounded supplementary search of bioRxiv and medRxiv will identify publicly available preprints meeting the same substantive PCC criteria. They enter a separately labelled frontier register, with separate flow counts and denominators, and are not pooled with peer-reviewed evidence.

Publication status is checked at the final update, and preprint/published versions are linked and merged appropriately. Clearly labelled public preprints may be cited under current Frontiers guidance [14]; this does not imply access to unpublished private results.

D.4 Date, language and retrieval

Searches run from database inception to their actual execution dates. No search date restriction is applied. The search-stage limits are those of amendment PRE-006 (E): the species and age limits in PubMed, the publication-type limits in all three databases and, by the team decision of 5 October 2026, an English-language limit and a journal-article document-type limit in all three databases. Reports in other languages or formats that reach the team through citation chasing, Google Scholar or the preprint supplement are still assessed: English and Chinese full texts are read directly; other languages use translation with independent human checking of eligibility and key data. The language limit is reported as a limitation. Unresolved language or eligibility information remains awaiting classification rather than being automatically scientifically excluded.

Institutional access and lawful public copies will be used for full texts. A sought but unobtained report is “not retrieved,” not a scientific full-text exclusion. This protocol does not plan author outreach; no contact is sent without separate authorization. Retrieval attempts and unresolved reports are logged.

E. Information sources and search strategy

Under amendment PRE-006 (5 October 2026), the bibliographic databases are PubMed, Web of Science Core Collection and Scopus, and Google Scholar is searched as a documented supplementary source rather than as a core database. PubMed is reported as PubMed rather than implying a separate MEDLINE-platform search. The search lead (D) runs Web of Science and Scopus in their web interfaces and exports from there; application-programming-interface routes were abandoned. Exact collections, platforms, interfaces and coverage will be recorded. Embase and SPORTDiscus, planned up to version 3.0, were withdrawn for licence and feasibility reasons; the resulting limitation is stated below.

The planned search is a single route, **E AND I AND T**, where E represents exercise, I immune concepts and T post-exercise or acute-bout wording. Controlled vocabulary and free-text fields are adapted to each platform. The immune block covers immune cells, mediators, function and mucosal immune terms through a list of named immune concepts rather than the **immun*** truncation, so that method words such as immunoassay, immunoblot or immunofluorescence do not by themselves count as immune concepts; the bare words **complement** and **cytotoxic*** are searched in titles, with specific phrases in abstracts. Infection outcomes are not search concepts, because an infection outcome alone does not meet the immune-link rule (C), and infection outcomes are charted whenever an eligible study reports them. The exercise block includes relevant training, competition and sport terminology; generic exercise words (for example **sport***, **train***, **competition***, **race***, **endurance** and **aerobic**) are searched in titles only, and the bare term **run** and the loose pairing of **physical** with **activit*** are not used. The former omics route (E AND O AND T) is dropped: under the immune-link rule an eligible omics study must have an identifiable immune anchor, so E AND O AND I AND T is a subset of E AND I AND T, and none of 61 positive seeds was retrieved only by the omics route. The omics vocabulary (transcriptomics, RNA/non-coding RNA, proteomics, metabolomics, lipidomics, epigenetics and multi-omics) is kept for charting and tagging. Web of Science and Scopus are searched in the title, abstract and author-keyword fields rather than Topic or TITLE-ABS-KEY, provided that D’s test on 10 seeds loses no seed; otherwise the broad fields are used. The search is by construct, not by label: “precision,” “biomarker,” “open window” and “recovery” are not mandatory search concepts, and the organizing window is applied at screening. The phrase “open window” is unsuitable as a filter—a PubMed title/abstract query combining it with exercise and immune terms returned

18 records for 2018–2026 at the 4 October 2026 re-check, and original studies rarely use it—so open-window-type terms may be added only as OR supplements or used for citation chasing, never as AND terms.

Version 3.0 applied all eligibility rules at screening rather than in the query. Amendment PRE-006 moves part of three rules into the query. (a) In PubMed, animal-only records are removed with NOT (animals[mh] NOT humans[mh]), the form used by Cochrane and CADTH, which keeps records not yet indexed with MeSH. (b) Reviews, editorials, letters, comments and case reports are removed by publication type in PubMed, and the equivalent document types are excluded in Web of Science (Review, Editorial Material, Letter, Correction) and Scopus (re, ed, le, no, er, sh, cr). (c) In PubMed, child-only records are removed with NOT ((infant[mh] OR child[mh] OR adolescent[mh]) NOT adult[mh]), which keeps mixed-age and unindexed records. (d) By the team decision of 5 October 2026, taken after D's first Web of Science and Scopus runs, the core routes are further limited to English-language records and to journal articles: PubMed AND english[la] NOT preprint[pt] (preprints stay in the supplementary preprint search), Web of Science DT=(Article) AND LA=(English) (records flagged Article and Early Access remain) and Scopus DOCTYPE(ar OR ip) AND LANGUAGE(english) (articles in press kept). Conference papers, meeting abstracts, book chapters and data papers are therefore dropped at search, consistent with FT01 and the title-stage closed list. These limits implement part of FT02 (animal-only studies), FT01 (source type) and FT03 (children only) in the query. The codes and their hierarchy are unchanged and apply at screening to every record the limits do not remove; Web of Science and Scopus have no species or age tag, so FT02 and FT03 remain screening rules for their records. Reviews removed by limit (b) are not screened; they are kept as a logged list of citation-chasing sources.

Amendment PRE-003 (4 October 2026) made T a required concept instead of a diagnostic block; T expresses that measurements follow an identifiable bout and does not encode the 0–72 hour window, which is still applied at screening. In PubMed parser validation on 4 October 2026, the construct-at-screening query E AND (I OR O) returned 255,019 records, about 8,500 person-hours of two-reviewer title/abstract screening before records from the other databases (assumed to add 1.5–2.5 times the PubMed volume before deduplication), whereas requiring T returned about 15,900 records in the decision test and 20,090 records with the finalised v0.7 T block, retrieving all 10 positive PubMed seeds in both runs. That evidence is limited: only 10 positive seeds were tested, the proximity form that recovered one missed seed was designed after the miss, and T is expected to miss reports that give timing only numerically (for example “at 1 h and 24 h”), so the trade-off is disclosed as recall given up for feasibility. Mitigations are a positive seed set expanded to at least 30 reports, including at least five that give timing only numerically, re-tested in the pilot; a T-free sensitivity run during the pilot, in which a random sample of records retrieved only without T is screened; and backward and forward citation chasing of all included reports and of the five open-window reviews named as citation-chasing sources (Peake et al. [15], Campbell and Turner [1], Simpson et al. [16], Xie et al. [17] and Shi et al. [18]).

Strategy v0.9, which implements PRE-006, rests on PubMed diagnostic counts of 5 October 2026; these are diagnostics, not formal searches. The v0.7 union of both routes returned 20,090 records. Adding the animal-only exclusion, title-only generic exercise words and the publication-type exclusion (v0.8) gave 11,448; adding the child-only exclusion, the removal of run and of the physical-activity pairing, the named immune list and the title restriction of complement and cytotoxic* gave 9,577; removing four generic timing phrases from T (“time course,” “time-course,” “pre and post,” “baseline and post”) gave 8,909; keeping only the route E AND I AND T gave 7,733; and adding recall terms to T (unhyphenated preexercise, postrace and prerace; “hr after” and “hr post” forms; “h,” “min” and “hours of recovery”; “acute effects”; and the race and event nouns marathon, ultramarathon, triathlon* and ironman) gave 8,013 records for the v0.9 query with the three NOT limits (run at 2026-10-05T13:03:39Z; SHA-256 prefix of the query e30ec141aac4).

With the English-language limit and the removal of preprints from the core route (decision of 5 October 2026), the final v0.9 PubMed route returns 7,846 records (run at 2026-10-05T13:23:01Z; SHA-256 prefix 6ad15420fe1f); the only seed that drops out is POS-S3, a preprint that belongs to the supplementary preprint search. The final query retrieved all 10 original positive seeds and 50 of the 51 expanded positive seeds; the missed seed was also missed by v0.7, and the six v0.7 misses that were recovered depend on recall terms chosen after those misses, so this is not an independent recall estimate. Relative to v0.7, the only seed-list records lost were the youth boundary seed, removed by the child-only exclusion as intended, and one record advanced in the 50-record pilot whose immune match was a method word. For comparison, 23 published reviews in the field identified a median of 2,360 records from databases (broad multi-family and scoping reviews: median 4,690, maximum 10,976; 03_search/strategy_review_2026-10-05.md). The projected unique volume across PubMed, Web of Science, Scopus and the Google Scholar supplement is roughly 9,000–13,000 records.

Full working strategies (v0.9) and a search-log template are in 03_search. The PubMed v0.9 strings have been run only as diagnostics, and the Web of Science and Scopus translations were validated on 5 October 2026 by D's seed test on the platforms (every indexed seed retrieved with the narrowed fields) and run with those fields. A's PRESS sign-off of 4 October 2026 applied to strategy v0.7 and is void under PRE-006; A re-reviewed v0.9 under PRESS [11] on 5 October 2026 and accepted it without changes, so the exports of 5 October 2026 (PubMed 7,846; Web of Science 13,242; Scopus 11,065 records, all with abstracts) are the formal database searches; the Google Scholar supplement is deferred until after deduplication and title screening. PRESS revisions and the resolution of comments are retained. The PubMed export of 5 October 2026 under v0.7 and D's first Web of Science export batch for the former omics route are superseded; they are kept as artefacts and are not PRISMA counts. Known eligible and boundary examples will test strategy behavior; the v3 seed list includes open-window-centred original studies of conventional indices (for example salivary IgA, natural-killer-cell and lymphocyte redistribution responses after prolonged exercise) and boundary cases (immediate-only sampling, sampling only beyond 72 hours, youth cohorts and resting training studies). Retrieval of a few selected positive seeds is a debugging check, not an estimate of overall recall. A seed not retrieved will be investigated; strategy changes must retain a conceptual rationale rather than simply fitting known papers.

Google Scholar is searched as a supplementary source. Because Google Scholar limits queries to 256 characters and does not support truncation, a small set of short query strings is used; the strings are drafted in the strategy file in 03_search, reviewed under PRESS with the rest of v0.9 and recorded verbatim. For each string, the first 200 results sorted by relevance are examined, and the search date, the number of results screened and the stopping rule are logged. Records found enter the same two-stage screening (F) and are reported under PRISMA-S [10] as a supplementary source.

Withdrawing Embase and SPORTDiscus limits recall. Published comparisons show that Embase adds unique records in biomedical reviews; Bramer and colleagues found that MEDLINE, Embase, Web of Science and Google Scholar together were needed to reach about 98% recall [ref to add]. SPORTDiscus indexes sport-science journals that are not all covered by Scopus or Web of Science. The search-stage limits add a smaller risk: PubMed records misindexed as animal-only, child-only or review, and Web of Science or Scopus records with an incorrect document type, are not screened. The English-language and article-only limits exclude non-English reports and non-article formats at search; exercise-immunology reports in other languages exist, so this is a recall limitation that is disclosed and only partly offset by citation chasing, which is not language-limited. Mitigations are backward and forward citation chasing of all included reports and of the five open-window reviews, the Google Scholar supplement and the expanded positive seed set; the limitation will be reported in the review.

Backward references and forward citations of included reports and relevant neighboring reviews

will be checked, including forward citation tracking of Peake et al. [15] and Campbell and Turner [1] before submission. Registered and ongoing reviews in OSF Registries, PROSPERO, INPLASY, Research Registry and Zenodo, together with adjacent recent reviews, will inform the novelty assessment; the 4 October 2026 registry scan is pre-archive reconnaissance and will be repeated before submission. Supplemental preprint searches will record sites, exact queries, dates, filters, results examined and stopping rules. Citation chasing has a defined sweep and documented closure/update; it is not silently unlimited browsing.

For every run, preserve the exact query, database/platform, run date and time, limits, hits, export counts, file format, immutable raw export and checksum. Record search amendments, supplementary routes and deduplication decisions under PRISMA-S. The target is to update searches within 30 days before submission; this is a project target, not a journal rule. New exports do not overwrite earlier exports.

Records are deduplicated using identifiers followed by normalized title/author/year checks and human confirmation of uncertain matches. Distinct reports of the same study or cohort remain linked, rather than being discarded as bibliographic duplicates. Study, report, cohort and preprint-version relationships are preserved. No formal yields are available at protocol preparation.

F. Selection and calibration

Screening has two record-level stages before full text (amendment PRE-005). At stage 1, two screeners independently screen the titles of **all** deduplicated records, seeing only the title, source, year, document type and identifiers. A record may be excluded at this stage only for a reason on a closed list that is evident from the title or document type: review, editorial, protocol or conference abstract (mapped to FT01); animal-only or in-vitro-only study (FT02); a cohort clearly under 18 years (FT03); no evident exercise or physical-activity context (FT04); or no evident immune, inflammatory or omics content (FT07). Every other record advances, including any record whose title is silent on a criterion. Consensus at stage 1 is automatic: a record advances if either reviewer advances it and is excluded only when both exclude it; conflicts are logged, not adjudicated. Abstracts are then completed for the retained records, taken in order from database exports that already contain them, from PubMed by PMID (or by DOI converted to a PMID), from the source database's full record and, failing these, by manual retrieval; a record without an obtainable abstract is screened on its title and other available text, flagged as such, and never excluded for that reason. At stage 2, two screeners independently screen the titles and abstracts of **all** retained records under the version 3 rules, and the human reviewers later screen **all** sought full texts (under PRE-008, the confirmed validation layer; under PRE-010, in override mode on AI pre-fills). At title/abstract stage, missing timing or methodological detail normally proceeds to full text rather than being guessed. Under amendment PRE-007 (6 October 2026), the two independent first-pass screeners at stages 1 and 2 are AI stand-ins with different prompt orientations (inclusion-oriented, defaulting to advance; and closed-list-strict), each reading every record individually, and every record on which they disagree receives a blind third AI read recorded separately. The stand-in passes are subject to mandatory human verification: reviewers B and C independently confirm every AI-advanced record, every conflict left without a third read and every awaiting-classification record, and screen a seeded 10% random sample of agreed AI exclusions with the AI columns hidden (seed 20261005); the human decisions are the decisions of record, and an advance found inside the sample triggers re-screening of that record type. Agreed AI exclusions outside the sample are listed and reported separately in the PRISMA-ScR flow as AI exclusions with 10% sample verification. A 100-record random title-pilot sample (seed 20261005) is issued to B and C as a blind calibration set with the AI columns hidden, and the AI-versus-human comparison on that sample is reported as the stand-in reliability check. The manuscript discloses the design and the models in Methods and Acknowledgments. The split was adopted when the five-database volume under strategy v0.7 was estimated at 60,000–90,000

records, and in response to uneven off-campus access to abstracts through database interfaces; title screening is about three to four times faster, and recall is protected by the narrow closed list and the advance-wins rule. Amendment PRE-006 later reduced the projected unique volume to roughly 9,000–13,000 records (E); the two stages are retained.

Under amendment PRE-008, the stage-2 AI advances are triaged before human verification. Two model families independently extract five abstract-level elements with verbatim quotes: exercise exposure (E1), post-exercise sampling (E2), immune marker (E3), population (E4) and the validation-readiness element (E5; C). Family disagreements, and records whose E5 rested only on repeated monitoring, receive a blind adjudicating read under the stricter E5 definition. The final elements place each record in one layer: X (a core element absent: no exercise exposure, post-exercise sample or immune marker, or a population under 18, clinical or non-human), V (core present, E5 present), M (core present, E5 absent) or U (E5 unclear). B and C independently confirm E5 and E4 for every V record with the AI quotes visible, decide every U record, and verify seeded 10% samples of M and X (seed 20261005) and of adjudicated records (seed 20261008); D resolves disagreements. Only confirmed V records proceed to full text and extraction; M is charted from abstract-level elements only; X is reported as AI exclusions with sample verification. Stage-1 verification under PRE-007 is unchanged. Before full-text screening, B and C complete the full-text boundary exercises (`04_screening/fulltext_boundary_exercises.md`) as the required calibration. At full text, B and C also record whether the validation-readiness element is confirmed; a V report whose element is not confirmed is kept and charted fully, flagged, and counted in the flow and in the map, not excluded for that reason.

Under amendment PRE-010 (8 October 2026), full-text screening of the confirmed V layer uses AI pre-fill with human override. Two AI stand-ins from different model families (GPT-6 Sol, reasoning effort high; GLM-5.3, thinking on) each read every retrieved full text and fill the full-text fields independently: disposition under the fixed FT01–FT08 hierarchy, age-rule check and validation-element confirmation (v1.2 definitions), with page-referenced supporting passages and quotes of at most 25 words. B's workbook carries the GPT-6 Sol read and C's the GLM-5.3 read; the other family's read stays hidden. B and C read every full text and overwrite what they disagree with; unchanged cells are their decisions of record, and a hidden pre-fill copy gives each reviewer's override rate. Disagreements between their final values are discussed and, if unresolved, adjudicated by D. B and C each pass the 15 full-text boundary exercises (at least 12 of 15) before the override phase. Reports not retrieved stay NOT_RETRIEVED and are never excluded. Methods report the design, models, prompt version and override rates; that screening was not independent from scratch is a limitation.

After the v3.0 protocol has been archived (J) and before formal screening, a saved random procedure selects 50 records from a frozen deduplicated draft-search pool; the pool is generated after the archive release and retained with its version. The pool version, eligible sampling universe, random seed, selected IDs and reviewer decisions are retained. The project criterion is at least 80% raw agreement plus resolution of all conceptual disagreements; this threshold is not attributed to JBI. Cohen's kappa is reported descriptively with prevalence limitations. If the criterion is unmet, the manual is clarified and a fresh pilot is performed. If the pilot changes an eligibility rule, the protocol is re-released with justification before formal searching (J). The 50-record pilot does not confer final study eligibility. Under version 3.1, a 100-record title pilot drawn from the deduplicated formal union with a saved seed must reach at least 80% raw agreement on the binary title disposition (advance versus exclude) before formal stage 1, and, because version 3.1 changed the screening codes, a 25-record title/abstract re-calibration drawn from the stage-1 retained set, judged by the original criterion, precedes formal stage 2. Full-text boundary exercises follow before formal full-text screening.

At stage 2 and at full text, disagreements are discussed; unresolved decisions go to a predesignated adjudicator whose name must be supplied before formal screening. Changes to an eligibility

interpretation are dated and applied to previously screened records when relevant. Operational records retain separate independent decisions and the consensus decision.

The screening manual defines this fixed primary exclusion hierarchy: FT01 ineligible source type; FT02 not original human research; FT03 ineligible population, including cohorts stated to include participants under 18 years without a separable adult stratum (age rule in D.1); FT04 no qualifying identifiable exercise context, where any intensity, duration or mode qualifies and the code applies only when no identifiable bout, or for B no qualifying repeated-bout relation, can be established; FT05 no post-cessation sample, where sampling only beyond 72 hours is not a reason for this code because the organizing window is a charting frame; FT06 no baseline or suitable comparator; FT07 no eligible immune link; FT08 inseparable mixed intervention. Only the first established scientific reason is assigned at full-text exclusion; secondary concerns and supporting passages remain in notes. Unresolved information is not treated as an established failure. Duplicate reports/versions, not-retrieved reports and awaiting-classification records have administrative statuses outside the scientific exclusion hierarchy. So does RETAIN_BACKGROUND (amendment PRE-004), assigned at full text to background reports as defined in D.2: it carries no exclusion code (the failed dimension is kept as a secondary note), appears as a separate line in the flow diagram and places the report in a background register outside every original-study denominator. The actual flow will reconcile identification, duplicates, title screening (with stage-1 exclusions by closed-list reason), title/abstract screening, validation-readiness triage (PRE-008; X, M and V layers), sought/retrieved reports, exclusions and included reports/cohorts, with the frontier register separate. Counts remain unavailable, not zero, until work is performed.

G. Data charting

G.1 Pilot, structure and verification

After the archive release, ten purposively chosen reports will calibrate charting across exercise designs, conventional assays, cell/biofluid omics, early samples, repeated-bout monitoring, and timing and population boundary cases, including an adolescent cohort used as a population boundary case. These reports are training material, not ten included studies, and may include cases ultimately excluded. Their source, selection rationale and unresolved questions are recorded in the pilot manifest. Pilot refinements are versioned before full extraction. The charting pilot was run on 5 October 2026 with AI agents standing in for both extractors; the team accepted it as calibration of rule clarity, not as human inter-rater agreement. Its findings, chiefly undefined row units, led to amendment PRE-004 (L). The reduced human calibration round on the revised rules is folded into the first five V-layer reports extracted, one per validation sub-type plus one background report (decision by A, 8 October 2026).

Under amendment PRE-009, extraction is AI-assisted with human verification. Two AI stand-ins from different model families each extract every confirmed V-layer report, with supplements, into the per-report form of the extraction template, with source locations and without screening fields or each other's output. D checks and loads one stand-in's output into B's workbook and the other's into C's. B and C each check every cell against the source, confirming or overwriting it and adding or deleting rows as the source requires; unchanged cells are their values of record. A hidden copy of the pre-fill gives each reviewer's override rate. D compares the two reviewers' final values and resolves disagreements against source passages. Methods report the design as AI-assisted extraction with human verification, with agreement between the reviewers' final values and the override rates; that extraction was not independent from scratch is a limitation. The v3 dictionary, template and manual in 05_extraction define the operational schema; the Excel workbooks used for screening, extraction and appraisal are generated from those JSON templates by `scripts/build_workbooks.py`, and CSV exports are versioned in git. Data use linked report, study and cohort tables with long-form assay/marker/time/comparison records. Raw independent extractions and reconciled data are retained separately. Background reports (D.2) are charted

by one AI stand-in and verified by one person, limited to bibliographic data, design, population, assay platforms and sample sizes, without per-marker results or precision-domain records.

G.2 Charting domains

- **Provenance:** Identifiers, publication/version status, correction/retraction checks and dates, study/cohort links, source locations, extractor and decision history.
- **Participants:** Eligibility and health definitions, infection screening, recruitment/analysis numbers, age range and separability of the adult stratum, sex, training status, older-adult subgroup, per-group/time/platform participant numbers and within-person pairing.
- **Exposure:** Design, exercise mode, duration, absolute and relative intensity metrics and values as reported, comparator, training background, previous load, co-interventions, environmental conditions and recovery schedule.
- **Sampling:** Exact time and time origin, and position relative to the organizing window (within window, outside-window recovery, pre-exercise baseline or matched control time); matrix/immune compartment; collection, processing, storage, delay/freeze–thaw; circadian timing; plasma-volume/hydration correction; urinary dilution adjustment; salivary flow/secretion rate; cell composition and normalization.
- **Measurements:** Marker family (A); original and standardized marker names/identifiers, modality, targeted/untargeted assay, platform, units, detection/quantification limits, quality control, batch effects, molecular-identification confidence, bulk/single-cell structure and integration method.
- **Results:** Relevant contrasts, direction and magnitude, effect estimates and reported CI/SE or other uncertainty, multiplicity control, missing data, quality-control exclusions and measured/adjusted confounders. Relevant null and discordant findings are retained.
- **Author open-window interpretation:** Whether the report uses the open-window term, invokes immunosuppression or susceptibility without the term, interprets changes as redistribution, or offers no window interpretation, with the source passage; recorded descriptively and never used for eligibility or weighting.
- **Precision domains:** All seven domains in C are charted separately: analytical reliability; temporal validity; immune specificity; individualization; functional/clinical linkage; independent validation and use; and feasibility. Subfields include personal baseline, within-person variation, test–retest/repeated-bout reliability, heterogeneity analyses, subgroup/individual model specification, validation type, participant-level separation, calibration/discrimination, thresholds and decision/implementation evidence where available.
- **Functions and outcomes:** Direct assays versus computed annotation; co-measurement versus tested association; outcome definition/ascertainment, symptom versus clinical versus pathogen outcome, biomarker timing relative to onset and follow-up duration.

For large omics datasets, chart one summary record of the tested universe per analysis and every immune candidate named in the main text, main tables or figure legends, regardless of significance; candidates prespecified in the original aims or methods are always charted, wherever their results appear. Features reported only in supplementary result tables are summarized by counts and locations rather than charted one by one, and unreported rows are never invented. Record any candidate-selection rule and unavailable supplementary data. Preserve result units and avoid selecting only significant features. Unreported effect sizes are NR. Values shown only in figures are not digitized: the direction and statistical significance are charted from the figure, legend or text, the figure panel is recorded, and the value is NR and flagged as figure-only. Estimated values are never presented as directly reported data. Where reliable extraction is impossible, use NR or UNCLEAR with the reason.

Charting units and conventions are fixed by amendment PRE-004. A measurement record is one named analyte by one author-reported statistical test, plus the omics summary records above.

A sample-set record is one cohort by group or arm by matrix by time point as sampled; assay platforms are listed as a field, with separate records per platform only when platform-specific sample sizes are reported. Precision-domain records form one set of seven per report, cohort and marker family, with analyte-level records only where analyte-specific evidence exists. Recruited participants are counted before any exclusion; the analysed number is that for the primary immune outcome, other denominators are recorded in notes, and per-platform numbers are recorded separately where reported. Named analytes are assigned to a marker family by analyte, not by platform. Author open-window interpretations are coded from the authors' own voice (adopted rationale, discussion and conclusions), exhaustively, each with a source location. Publication status records peer-review and version status only; short-report formats are coded by their review status, and the article type is recorded with the version. Coded fields use controlled vocabularies, missing values follow a fixed not-applicable, not-reported, unclear order, and nothing is inferred from platform conventions, typical practice or figure icons. Since 8 October 2026, tested associations of a marker with an outcome or another marker are coded as positive, negative or no detected association; before a cohort is counted as independent, it is checked against the other charted cohorts for the same institution, overlapping recruitment and the same protocol or registry, with the result in the cohort notes. Two copy-only report fields record the validation sub-types that placed the report in V (from the triage) and the full-text confirmation of the validation element; the full-text field governs when they disagree.

Participant count, biological sample count, cell count and technical repeats are different fields. The donor is the independent unit for single-cell inference; cells do not replace participants. Model splits are assessed at the participant/cohort level, including feature selection inside training folds. Raw individual change scores without measurement/within-person noise assessment do not establish reproducible responder status.

G.3 Time representation

Exact exercise-to-sampling times are primary. Optional display bins are: pre-exercise baseline; matched control time; during exercise (ancillary); 0 to <30 minutes; 30 minutes to <3 hours; 3 to <24 hours; 24 to 72 hours inclusive; >72 hours; and NR. Bins up to and including 24 to 72 hours lie within the organizing window; >72 hours is outside-window recovery. Bins are mutually exclusive display categories, not eligibility thresholds. The numeric time is the earliest stated bound of a reported interval; an interval that straddles a bin boundary is assigned to the later bin and flagged; day-level timing is converted to nominal hours; and "immediately after" without a number is assigned to 0 to <30 minutes and flagged. Otherwise unknown timing is not imputed to a bin; uncertainty about exercise linkage may prevent main-map eligibility. Symptom/infection follow-up uses a separate time axis, with marker measurement before, concurrent with or after onset recorded explicitly.

H. Critical appraisal

Critical appraisal is prespecified to explain evidence limitations, not to create a clinical recommendation. Two reviewers independently apply tools matched to actual design, with discussion and adjudication. Current JBI tools are planned for randomized trials (2023 revision), quasi-experimental studies (2024), cohorts (2025), analytical cross-sectional studies (2026), and other eligible designs using the applicable current checklist [12]. Single-group pre/post studies are not treated as randomized trials. Crossover/repeated-measure limitations are charted explicitly. True diagnostic-accuracy questions use the current JBI diagnostic-accuracy tool. True individual-level future clinical-outcome or immune-response prediction-model development/validation studies additionally use PROBAST+AI (2025) [13]; a statistical model, molecular discovery classifier or simple association alone does not trigger that tool.

The exact official forms, versions, access dates and checksums will be archived before appraisal.

Native response categories, domains and supporting source passages are retained. We do not assume all JBI forms have identical item structure or force them into a common score. The manual in `05_extraction/critical_appraisal_manual.md` governs design selection and overlap.

Assay/omics reporting and the seven precision domains are recorded separately from design appraisal. They include analytical reliability, pre-analytical handling, normalization, cell composition, repeated-measure analysis, multiplicity and independent validation. These custom descriptors are not a validated risk-of-bias tool. There is no quality sum, quality-threshold exclusion, GRADE certainty assessment or numerical readiness ranking; domain evidence enters only the validation-readiness map as defined in A. Appraisal findings contextualize the maps, including unresolved reporting limitations.

I. Synthesis and planned outputs

Synthesis is descriptive, stratified and narrative. Independent cohorts are the primary count for evidence density; report counts and distinct studies are also reported with explicit linkage. Platform features, cells, assays or time points are never used as proxies for evidence strength. Denominators are specific to each question and include applicable/NR/UNCLEAR information as appropriate. A/B overlap is disclosed, and the two stream totals are not simply added.

Three primary evidence maps are planned:

1. **Open-window time by immune compartment and marker family:** Exact sampling coverage and optional time bins within and beyond the organizing window, by immune compartment or matrix, marker family, exercise context and population stratum.
2. **Per-marker validation-readiness map:** Marker or marker family by the seven precision domains, with conventional and omics candidates distinguishable; the descriptive profile defined in A, not a score or ranking.
3. **Cohort-level validation links:** Technical confirmation, repeated bouts, independent participants/cohorts, directly measured function, tested associations and clinical/prediction outcomes, with explicitly labelled gaps.

Under amendment PRE-008 the evidence base has two layers. The validation layer (V) holds the records confirmed by B and C as carrying a validation-readiness element (C); only these are screened at full text and extracted, and only they populate maps 2 and 3 and the main tables. The map layer (M) holds records with every core element but no validation-readiness element; they are charted from the abstract-level elements (exercise exposure, sampling, marker families, population, study type) without full text and are reported as a separately labelled abstract-level landscape, never merged with full-text denominators. In the PRISMA-ScR flow a validation-readiness triage box follows title/abstract screening and splits the AI advances into X (AI exclusions with 10% sample verification, by missing core element), M (a separate box, not sought at full text) and V (sought for full text).

The PRISMA-ScR flow is an additional process figure. It reports counts for each screening stage, with a separate box for title screening whose exclusions are given by closed-list reason and its FT code (F). Main tables describe study/cohort context, measurement/timing and limitations, individualization/validation evidence, and differences from adjacent reviews. Core A, support B and the frontier preprint register are visibly separated. Background sources do not enter original-study denominators. Counts of studies reporting a direction may be descriptive but are not a vote-counting estimate of effect or truth. Null, contradictory and unreported findings remain visible.

We will not pool effects, derive a diagnostic threshold, validate a marker, recommend return-to-training decisions, conclude that a candidate predicts infection from group differences alone, or adjudicate whether an open window exists. Discovery of a potentially homogeneous subset will be

documented as a future focused-review opportunity. Any later decision to add an effect synthesis requires an explicit protocol amendment and appropriate methods before analysis. The manuscript will retain the scoping methodology regardless of the submission portal's article-category label [14].

J. Registration, archiving, governance and transparency

The protocol is archived publicly rather than registered in a review registry. After the research team froze version 3.0, the public GitHub repository was tagged and the GitHub Release was archived automatically by Zenodo on 5 October 2026, which assigned a dated, immutable version DOI (10.5281/zenodo.23147973) and a concept DOI for the record series (10.5281/zenodo.23147972). The release preceded the 50-record screening pilot and the 10-report charting pilot, and formal database searching starts only after an archived release exists. Because the charting pilot changed charting rules, the team adopted two-stage screening and the database set and search strategy were revised, version 3.1 (amendments PRE-004, PRE-005 and PRE-006) is released with justification, as a new tag and version DOI under the same concept DOI, before formal searching; A's PRESS re-review of strategy v0.9 was completed on 5 October 2026 (accepted without changes). PROSPERO does not accept scoping reviews, and OSF Registries is no longer the planned route. The project has never been registered in any registry.

Pre-archive knowledge is disclosed rather than denied. An internal protocol (version 1.0, January 2026) with its own literature collision assessments, public reconnaissance and source verification on 2 October 2026 (Codex), and a collision re-check of the narrowed framing against published and registered reviews on 4 October 2026 (Claude Code; 02_preliminary/collision_recheck_2026-10-04.md) all precede the archive, and the two pilots are planned stages that will also precede formal searching. Their dates, scope and influence on decisions are recorded in the archive release record and the amendment log. The work is not described as wholly prospective from before any knowledge of the literature.

Substantive amendments record the previous rule, new rule, reason, date, work stage, information already known, authorizing team decision and need to revisit earlier records; after the archive release, each amendment also produces a new tagged Release and version DOI. Raw search exports and independent decisions remain immutable. The canonical protocol and execution appendices are versioned together.

This review uses published or lawfully accessible material and does not recruit participants or intervene in exercise. Institutional requirements for ethical review/exemption will be checked and recorded; no approval or exemption number is invented. The GitHub repository and Zenodo archive hold the protocol, manuals, templates, search strategies, dictionaries, decision metadata and shareable synthesis data under CC BY 4.0 for documents and MIT for scripts; copyrighted full texts and raw database exports are not redistributed contrary to their terms. Author names, affiliations, funding and competing interests are pending human declarations.

AI has supported public-source reconnaissance, evidence organization, consistency checking and drafting in this project: Codex desktop with delegated agents on 2–3 October 2026, and Claude Code (Anthropic) on 4 October 2026 for the collision re-check and the v3.0 revision, and on 5 October 2026 for the AI-assisted charting pilot, the PubMed diagnostics of strategy v0.9 and the drafting of amendments PRE-004, PRE-005 and PRE-006. Tool names, versions, models, sources, dates, tasks, prompt/output references and human verification are logged in 01_protocol/ai_use_log.json, and the manuscript will disclose tool name, version, model and source in both Methods and Acknowledgments [14]. Under PRE-007 and PRE-008, AI stand-ins, third reads and the adjudicating triage read (5–8 October 2026; logged in the same file) produce recommendations that humans verify as specified in F; under PRE-009 and PRE-010, AI stand-ins pre-fill the extraction (G.1) and the full-text screening fields (F), which B and C verify cell by

cell. AI is not an author, human independent reviewer or the human adjudicator. Named humans remain responsible for eligibility, data, appraisal, citations and conclusions. No AI-produced citation or numerical claim is accepted without source verification. Final disclosure follows the then-current journal and institutional requirements.

K. Roles and execution plan

Roles to name are: principal investigator; reviewers A and B; search lead; independent PRESS reviewer; adjudicator; data/version manager; and analysis/writing lead. A second human reviewer and access to PubMed, Web of Science Core Collection, Scopus, Google Scholar and full texts are available. Other roles may overlap where independence is preserved, but an adjudicator must be named before formal screening. Funding and competing interests cannot be inferred from the absence of information.

Screening, extraction and appraisal use Excel workbooks generated from the JSON templates in `04_screening` and `05_extraction` by `scripts/build_workbooks.py`: a read-only record master, one locked decision workbook per reviewer, dropdown-controlled codes, and identifier and gene-name columns stored as text. Each reviewer's workbook is locked at completion and its SHA-256 hash committed to git before the two files are merged; `scripts/merge_screening.py` performs the deduplication checks, merge, agreement, kappa and PRISMA counts, and CSV exports are versioned in git. Rayyan, Covidence or similar platforms are not used. A reference manager with exportable identifiers is still to be chosen; software and versions are recorded before execution.

Preparation is estimated at approximately 2–3 working weeks, depending on team availability: 3–5 working days to verify platforms and search strategies and to generate the workbooks; 2–3 working days to resolve team comments, freeze version 3.0 and publish the archived release; and 5–7 working days for reviewer training and the pilot screening and charting, with parallel work where feasible. If the pilots change a rule, a conditional re-release follows before formal searching. These are planning estimates, not a promised completion date. Full-review effort will be estimated after the 50-record screening pilot, the 100-record title pilot and the 10-report charting pilot, using actual yields and per-record effort. A provisional budget should distinguish two-person screening/extraction from single-person work and allow adjudication and update searches.

Execution order after the stage-2 AI pass (PRE-007, PRE-008): (1) D triages the 28 stage-2 advances added by the batch-235 re-screen with the same models and prompts; (2) A releases version 3.2 (tag, GitHub Release, Zenodo version DOI) after D's integration check; (3) B and C complete human verification (V confirmation, U decisions, the M, X and adjudication samples, the stage-1 and stage-2 exclusion samples and the 100-record blind title pilot), with D adjudicating; (4) A and D retrieve full texts of confirmed V records through `fulltext_requests/`; (5) A releases version 3.3 (PRE-009, PRE-010 and the PRE-004 update); B and C pass the full-text boundary exercises and screen V full texts in override mode on AI pre-fills (F); AI stand-ins pre-fill the extraction and B and C verify it (G.1), the first five V reports serving as the charting calibration, while M is charted from abstract-level elements; (6) synthesis.

The local execution plan and archive release record specify stage gates and deliverables. Formal stages remain pending until the required human work is completed; completing this document does not certify PRESS review, calibration or archiving.

L. Version history and decision status

Version 1.0 (January 2026) and associated build artifacts are preserved in the dated archive. Version 2.0 (2 October 2026) replaced the open-window-led title with candidate biomarkers for post-exercise precision monitoring; defined seven independent precision domains; separated acute A from strictly time-linked repeated-bout B; retained immediate samples without a 72-hour ceiling; expanded searching to inception without a language filter; removed mandatory

precision/biomarker/open-window terms; added cohort linkage, complete outcome/validation fields, design-appropriate descriptive appraisal and AI transparency; and removed unsupported novelty claims.

Version 3.0 (designed 4 October 2026; frozen and archived 5 October 2026) follows the collision re-check of 4 October 2026. It re-adopts the open window as the organizing construct that defines the 0–72 hour post-exercise window, the sampling time points and the marker families, without adjudicating whether it exists; defines the per-marker validation-readiness map as the deliverable; removes any exercise intensity, duration or mode threshold; restricts the population to adults aged 18 years or older, with older adults as a subgroup; retains Support B as the cross-bout monitoring layer; adds the author open-window interpretation tag, marker family and window-position fields; replaces OSF registration with a public GitHub repository and a Zenodo-archived Release, to be published after team freeze and before the pilots; specifies Excel workbooks generated from the JSON templates with scripted merging and agreement; adopts CC BY 4.0 for documents and MIT for scripts; and adds the Frontiers in Immunology submission requirements (PRISMA-ScR checklist and flow diagram, AI-use disclosure in Methods and Acknowledgments). Amendment PRE-003 (4 October 2026, decision by A; included in the v3.0 release) makes the post-exercise/acute-bout block T a required search concept (E AND (I OR O) AND T) on PubMed validation evidence, with an expanded seed set, a T-free pilot sensitivity run and citation chasing as mitigations, and records the PRESS conditions (no infection-outcome terms; **train*** retained pending the pilot test).

Version 3.1 (5 October 2026) implements amendment PRE-004 (decision by A), adopted after the 10-report charting pilot, in which AI agents stood in for both extractors and which the team accepted as calibration of rule clarity; most disagreements arose from undefined row units rather than misreading. It fixes the charting units for measurements, sample sets and precision-domain records (adding an optional marker-family column to the precision records); adds the marker family “mucosal antimicrobial protein”, sample roles for periodic training monitoring and resting habitual groups, and a candidate-selection value for statistically selected hits; converts matrix, exercise mode, training status, design and time unit to controlled lists; defines time-bin boundary rules, denominators, missing-value coding, author-interpretation coding and publication status for short reports; adds the administrative full-text status RETAIN_BACKGROUND, with background reports charted at reduced depth by one extractor and one checker; makes the adult age rule operational (explicit range or minimum of at least 18 years, or mean minus two standard deviations of at least 18 years; otherwise awaiting classification); and replaces figure digitization with direction-and-significance charting of figure-only results. Time-bin identifiers, the seven precision domains and the legacy field names are unchanged. Three optional proposals (association values for result direction, a per-contrast denominator field and study-family rules) are deferred. No formal search, screening or extraction had been performed when the amendment was adopted.

Version 3.1 also implements amendment PRE-005 (5 October 2026, decision by A), which replaces the single title/abstract stage with two-stage screening (F). Stage 1 screens titles only, with exclusion limited to a closed list of title-evident reasons mapped to FT01–FT04 and FT07, automatic advance-wins consensus and a 100-record title pilot (at least 80% raw agreement) before formal use. Abstracts are then completed for the retained records in a fixed source order, and records without an obtainable abstract are flagged rather than excluded. Stage 2 applies the unchanged version 3 title/abstract rules after a 25-record re-calibration. The PRISMA-ScR flow gains a title-screening box. The reason is feasibility: five-database volume was estimated at 60,000–90,000 records from the counts of 5 October 2026 (PubMed 20,090; exercise-omics (EO) route 6,669 in Web of Science and 6,159 in Scopus), abstracts are not uniformly available through database interfaces off campus, and title screening is about three to four times faster. Recall is protected by the narrow closed list and the advance-wins rule. No formal records had been screened.

Version 3.1 also implements amendment PRE-006 (5 October 2026, decision by A), “Database set, search-stage limits and strategy v0.9” (E). The databases are PubMed, Web of Science Core Collection and Scopus, with Google Scholar as a documented supplementary source; Embase and SPORTDiscus are withdrawn for licence and feasibility reasons, and the loss of recall is disclosed as a limitation with citation chasing, the Google Scholar supplement and the expanded seed set as mitigations. PubMed animal-only and child-only exclusions and publication-type or document-type limits in all three databases move part of FT02, FT03 and FT01 into the query; removed reviews become a logged citation-chasing list. Strategy v0.9 restricts generic exercise words to titles, replaces **immun*** with named immune concepts, restricts bare **complement** and **cytotoxic*** to titles, revises the T block and drops the omics route, leaving E AND I AND T; Web of Science and Scopus use title, abstract and author-keyword fields if D’s 10-seed test loses no seed. In PubMed diagnostics on 5 October 2026 the volume fell from 20,090 records (v0.7) to 8,013 (v0.9 with the three NOT limits) and 7,846 with the English-language and preprint limits, with no peer-reviewed positive seed retrievable by v0.7 lost. A’s PRESS sign-off of strategy v0.7 is void; A re-reviewed and accepted v0.9 on 5 October 2026. The PubMed v0.7 export of 5 October 2026 and D’s first Web of Science batch for the omics route are superseded artefacts, not PRISMA counts; the 50-record screening pilot, the 10-report charting pilot and the 200-record T-free sample remain valid calibration, and citation-chasing flags for records already in the pool are recomputed against the v0.9 pool. No formal records had been screened.

Amendment PRE-007 (6 October 2026, decision by A of 5 October 2026, registered retrospectively) replaces the v3.1 rule that AI makes no automatic exclusions, replaces no reviewer and shows no hints with AI-stand-in first-pass screening and mandatory human verification (F): two AI stand-ins with different prompt orientations screen every record of a stage, every disagreement receives a blind third AI read, and reviewers B and C independently verify all AI advances, all conflicts without a third read, all awaiting-classification records and a seeded 10% sample of agreed AI exclusions with AI columns hidden; agreed exclusions outside the sample are reported separately in the PRISMA-ScR flow. The reason is feasibility at the actual deduplicated volume (15,750 records; roughly 44 hours per person for dual human title screening alone), with recall protected by the advance-wins merge, the closed exclusion lists, third reads of every disagreement and the mandatory human verification. Stage-1 title screening was executed under this design on 5–6 October 2026 (15,750 records screened by both passes, 2,032 disagreements third-read, 7,606 advanced, 8,144 excluded with logged codes; `04_screening/formal_2026-10-05_v0.9/ai_stage1/`), and stage-2 title/abstract screening of the 8,447-record retained pool (7,606 advances plus 841 stage-1 conflict records retained for stage 2 under PRE-005) is in progress. PRE-007 was bundled into the version 3.2 release of 8 October 2026.

Version 3.2 (archived 8 October 2026; version DOI 10.5281/zenodo.23229246) bundles PRE-007 and amendment PRE-008 (8 October 2026, decision by A), “Validation-readiness triage of stage-2 advances with AI element extraction, GPT-6 Sol adjudication and layered human verification” (C, F, I). Stage-2 screening was completed on 7 October 2026: 9,277 records (the 8,447-record pool plus 830 stage-1 conflict records missed by an intermediate build) were screened, 4,326 advanced, 4,949 were excluded and 2 await classification. The 4,298 advances present when the triage was built were read by GLM-5.3-Flash (thinking off) and GPT-6 Luna (reasoning effort max) under prompt v1.1; 1,991 records were adjudicated by GPT-6 Sol (reasoning effort medium) under prompt v1.2, which tightened E5 after a spot check found crossover designs counted as repeated monitoring and background framing counted as outcome linkage. Final layers: V 577, M 3,020, X 697, U 4. Only V proceeds to full text. Methods will report the models, prompt versions and agreement statistics; the post-hoc narrowing of the full-text scope is a limitation. Version 3.2 is released before human verification and full-text screening start.

Amendment PRE-009 (8 October 2026, decision by A), “AI-assisted extraction with human verification” (G.1), replaces independent dual human extraction for the about 580 V-layer reports

because of workload: two AI stand-ins from different model families extract each report, B and C verify the pre-fill cell by cell with override rates recorded, and D adjudicates differences between their final values. The same decision updates PRE-004: association values for result direction are adopted, the shared-cohort check is adopted without the other study-family rules, the per-contrast denominator stays deferred, and the reduced charting calibration is folded into the first five V reports. Two copy-only fields record the triage validation sub-types and the full-text confirmation of the validation element; an unconfirmed V report is kept, charted and counted, not excluded. PRE-009 is released in version 3.3 (below). No V report had been screened at full text or extracted when it was adopted.

Version 3.3 (pending; to be released before full-text screening starts) bundles PRE-009, the PRE-004 implementation update and amendment PRE-010 (8 October 2026, decision by A), “AI pre-filled full-text screening with human override” (F). PRE-010 replaces dual independent human full-text screening of the 578 V-layer reports because of workload: GPT-6 Sol (reasoning effort high) pre-fills B’s full-text workbook and GLM-5.3 (thinking on) pre-fills C’s, each from its own reading of the full text; B and C read every report and overwrite what they disagree with, override rates are reported, and D adjudicates unresolved disagreements. The 15 full-text boundary exercises (pass rule 12 of 15) precede the override phase, and not-retrieved reports are never excluded. The PRISMA-ScR flow keeps its structure. No full-text decision existed when PRE-010 was adopted.

Still pending are named administrative declarations, the Google Scholar search strings (the supplement is deferred until after deduplication and title screening), the reduced human charting calibration (within the first five V reports extracted), the human 100-record blind title pilot and the 25-record title/abstract re-calibration, archived appraisal forms and local software configuration. Subsequent extraction and synthesis are distinct execution stages and have not been performed.

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